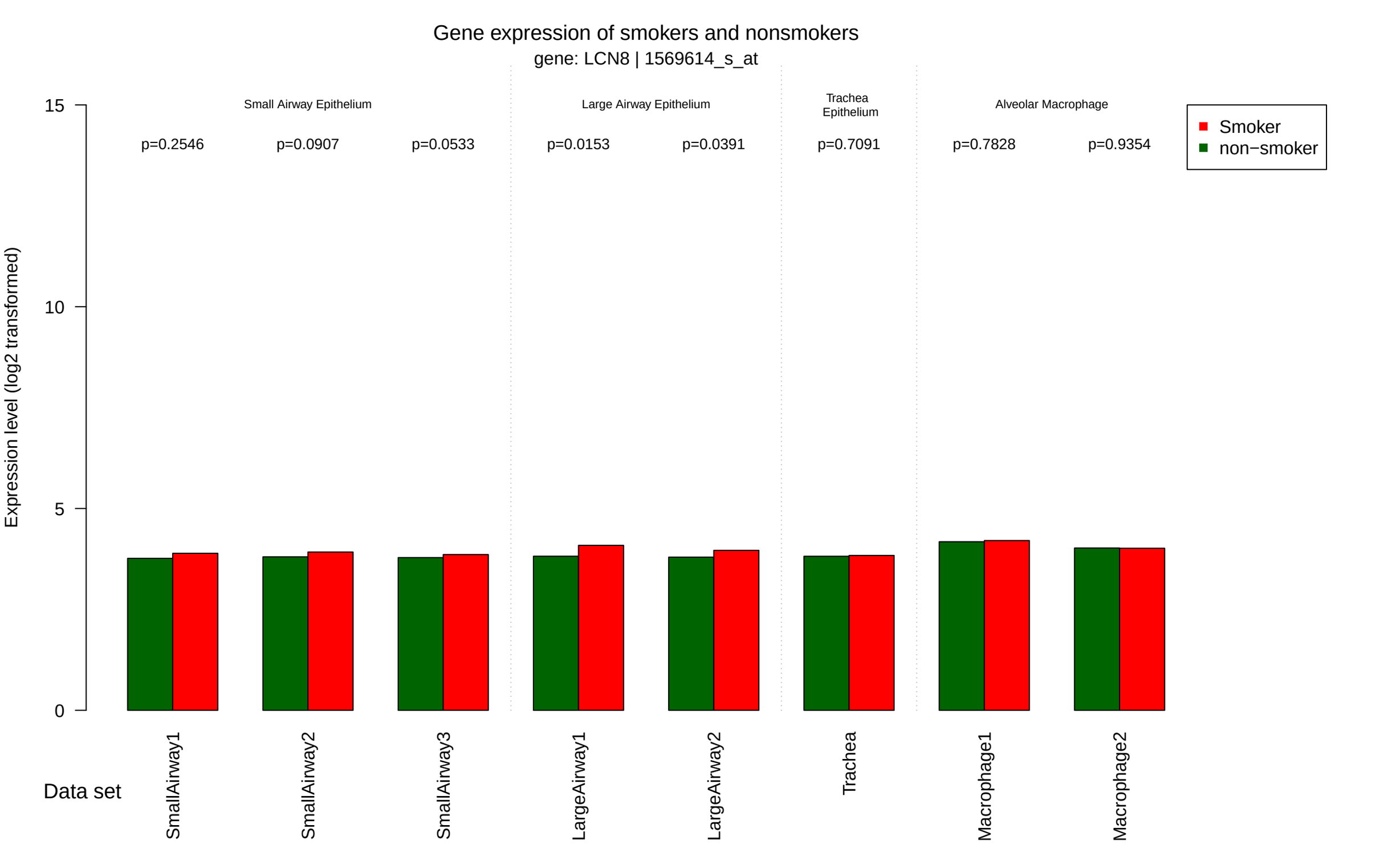
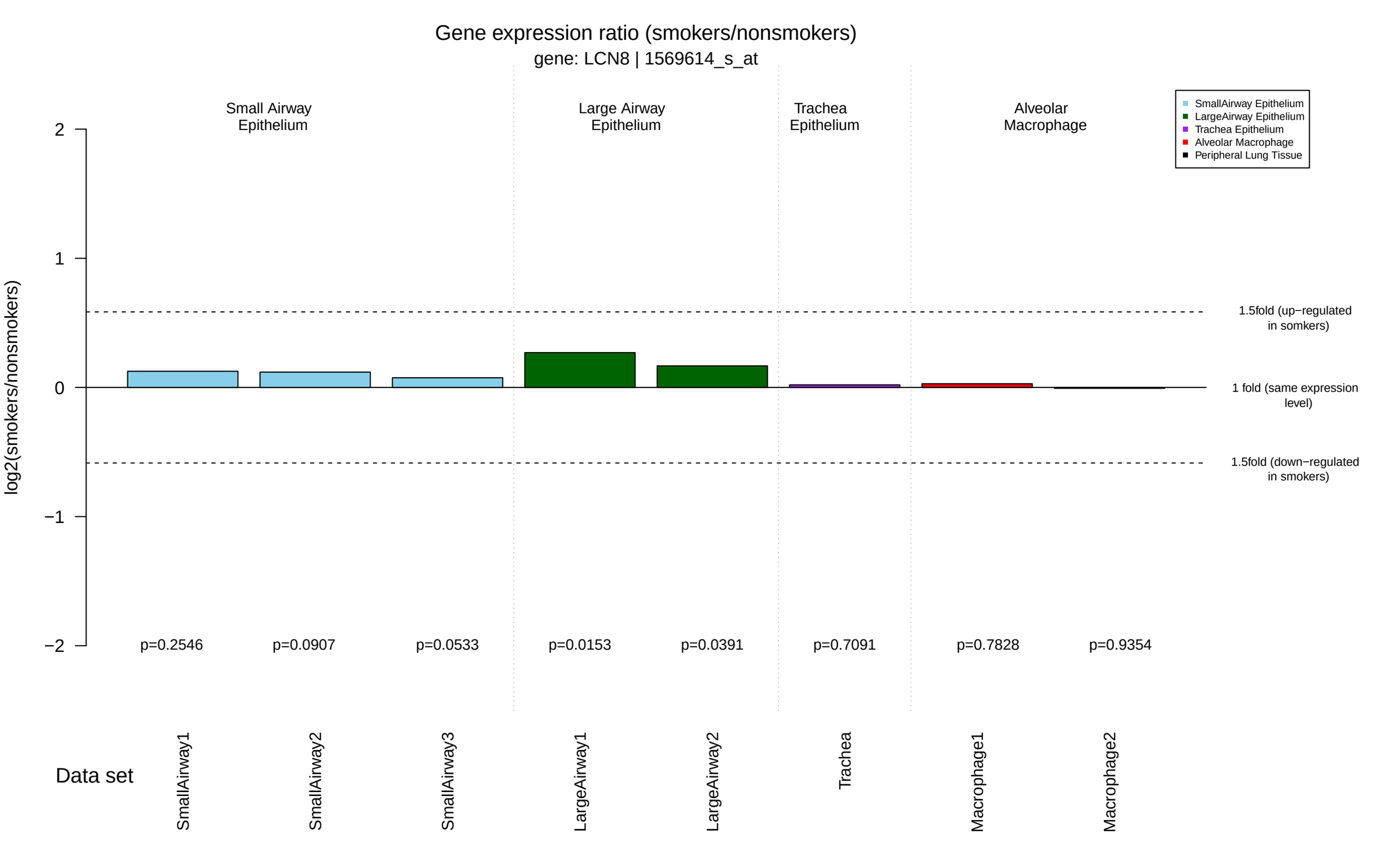
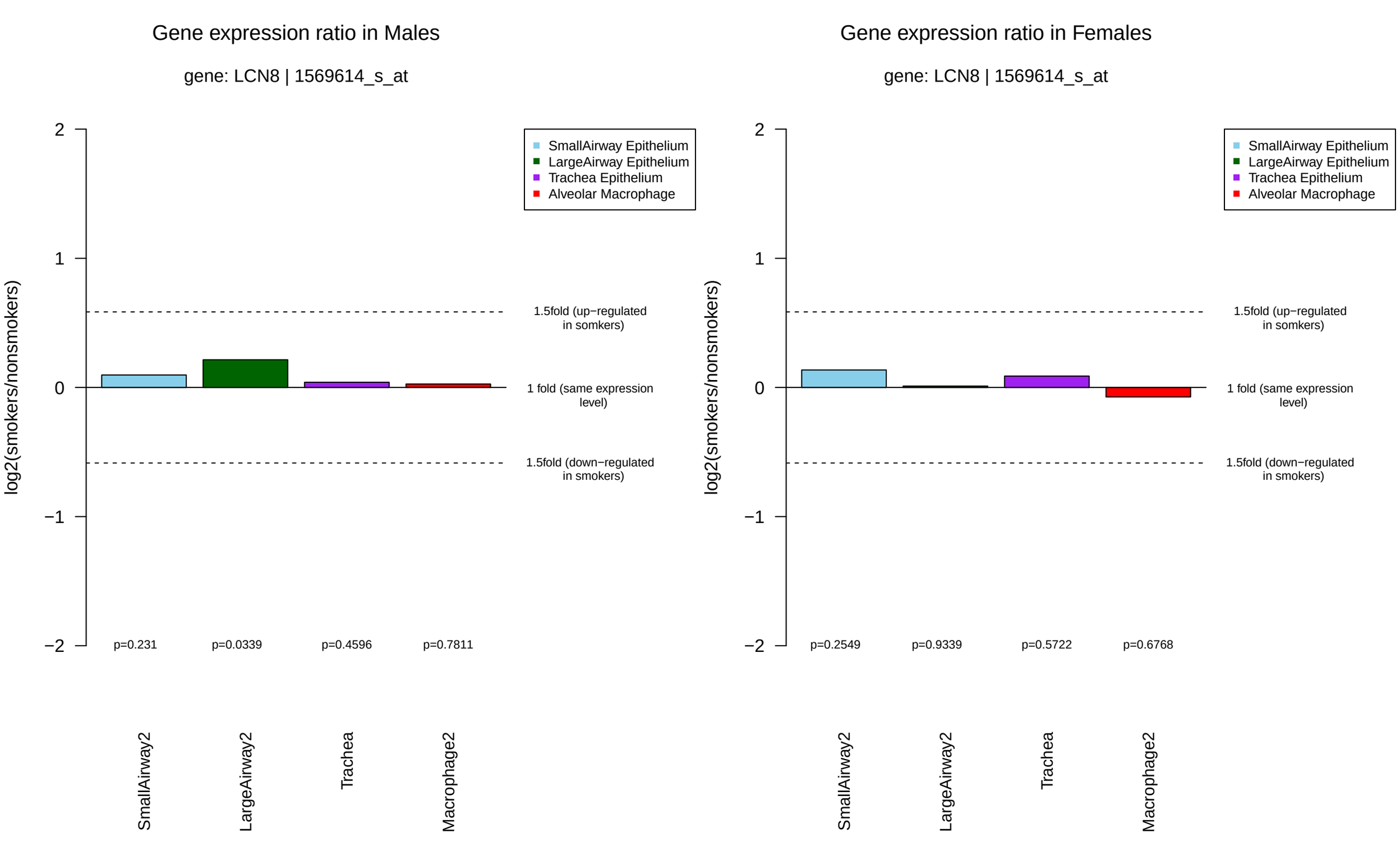
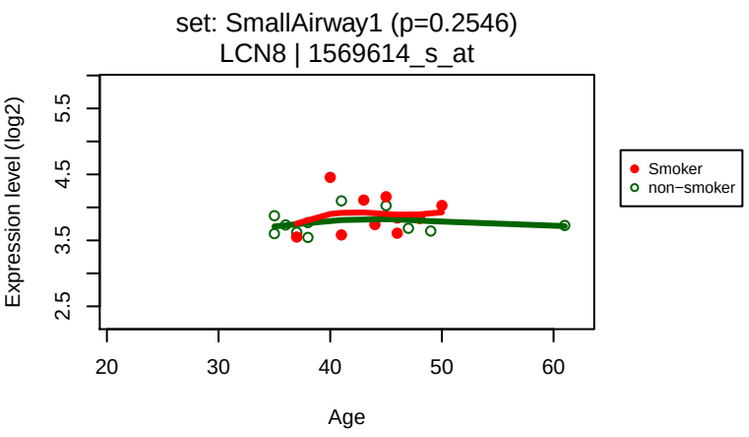
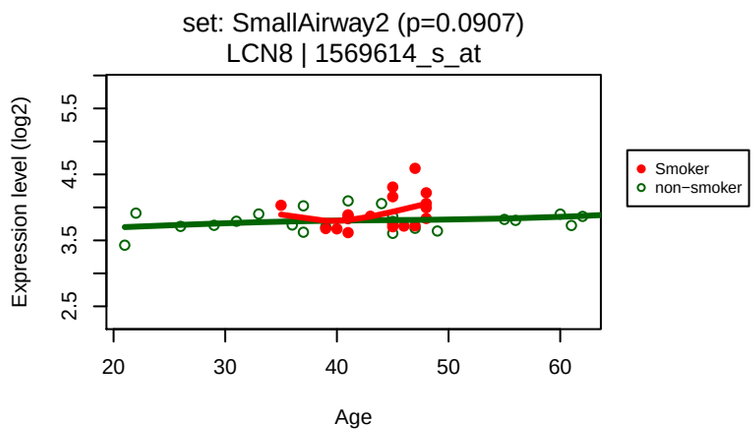
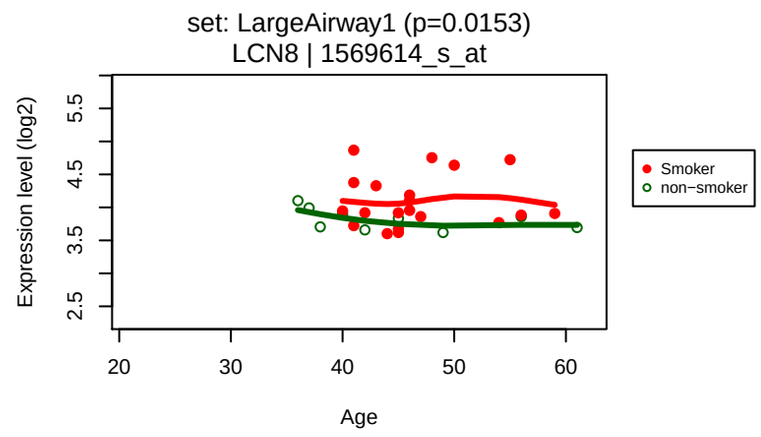

Gene expression of smokers and nonsmokers
gene: LCN8 | 1569614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2546
p=0.0907
p=0.0533
p=0.0153
p=0.0391
p=0.7091
p=0.7828
p=0.9354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

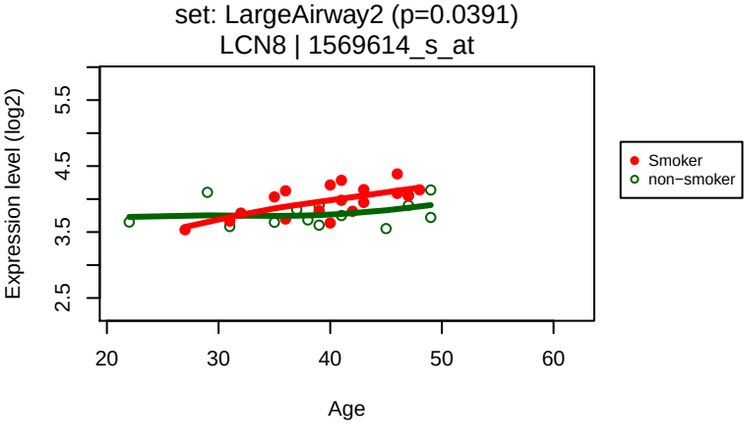

Gene expression ratio (smokers/nonsmokers)
gene: LCN8 | 1569614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2546
p=0.0907
p=0.0533
p=0.0153
p=0.0391
p=0.7091
p=0.7828
p=0.9354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

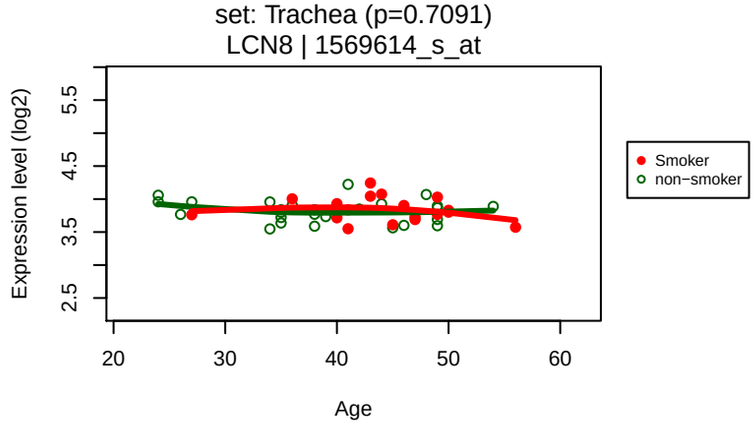

Gene expression ratio in Males
gene: LCN8 | 1569614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.231
p=0.0339
p=0.4596
p=0.7811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LCN8 | 1569614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2549
p=0.9339
p=0.5722
p=0.6768
SmallAirway2
LargeAirway2
Trachea
Macrophage2

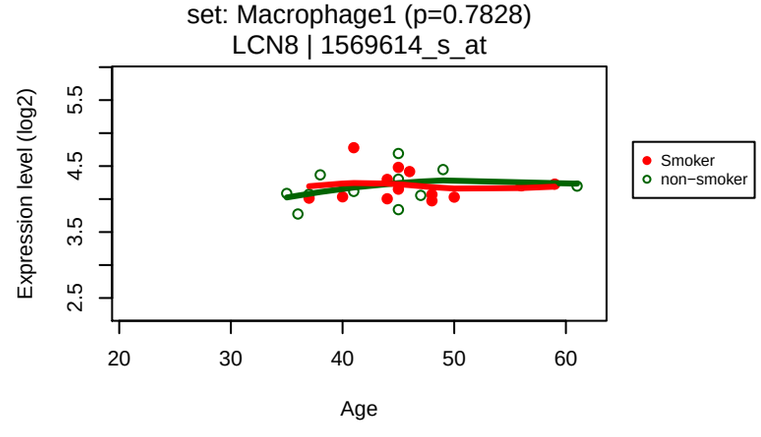

set: SmallAirway1 (p=0.2546)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker

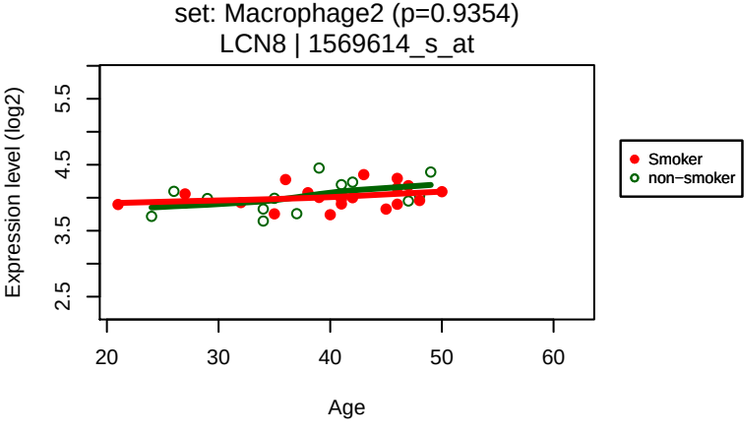

set: SmallAirway2 (p=0.0907)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker

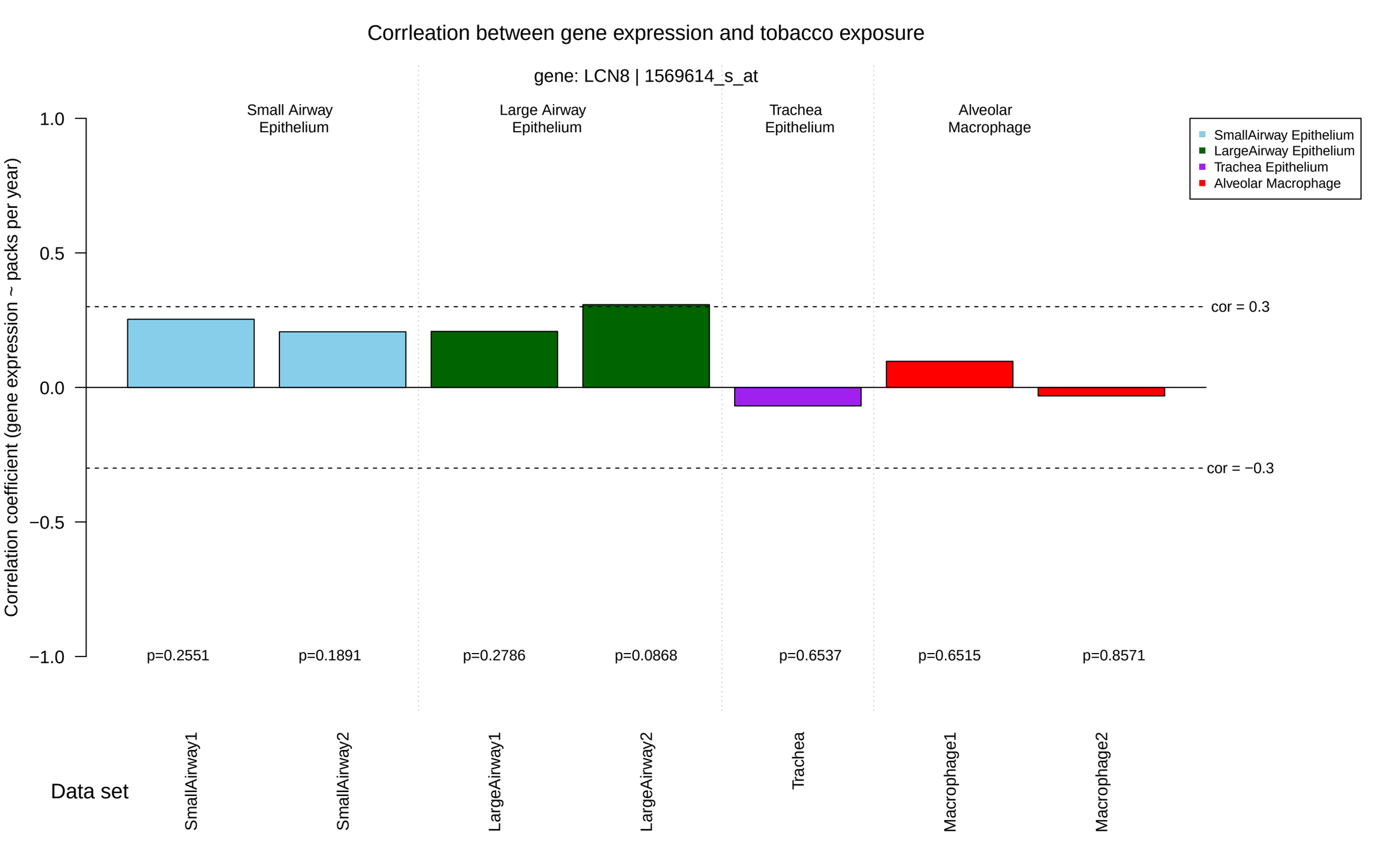

set: LargeAirway1 (p=0.0153)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0391)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7091)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7828)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9354)
LCN8 | 1569614_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LCN8 | 1569614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2551
p=0.1891
p=0.2786
p=0.0868
p=0.6537
p=0.6515
p=0.8571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2